A reversible splitting with nonpalindromic coefficients

Conjecture 00000006563

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On $\mathbb{R}^2$, take the two translation flows

$$A_t(x,y) = (x+t, y), \qquad B_t(x,y) = (x, y+t).$$

They obey the flow laws and have identity at $t = 0$. They are distinct: $A_1(0,0) = (1,0) \neq (0,1) = B_1(0,0)$. Consider the alternating splitting

$$S_h = A_{h/3} \circ B_h \circ A_{2h/3}.$$

Its coefficient list $(1/3, 1, 2/3)$ is not palindromic. Nevertheless direct substitution gives

$$S_h(x,y) = (x+h, y+h), \qquad S_{-h} \circ S_h = S_h \circ S_{-h} = \mathrm{id}\,.$$

Thus the actual map is time reversible for every real step size while coefficient mirroring fails. This refutes necessity, hence the asserted necessary-and-sufficient criterion.

Scope. Reversibility means the standard self-adjoint identity $S_{-h} = S_h^{-1}$. The problem imposes no noncommutativity or free-generator assumption and does not require reversibility uniformly over all possible underlying flows. In a free formal composition algebra, or under that additional universal requirement, the palindrome criterion is a different assertion. The witness here uses two distinct commuting flows, nonzero coefficients, and the actual specified splitting; no zero steps are inserted.

Lean correspondence. Lean defines the real translation maps, composition, coefficient list, and two-sided reversal identity. It proves both flow laws, the common identity map, distinctness, the exact translation formula, reversibility and nonpalindromicity. The final theorem combines the last two conclusions.

Reproduction. The standalone project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409` Run `sh verify.sh`; the Lean source prints the main results' axiom dependencies. No proof placeholders, native evaluator bypass, or custom axioms are used.

Source. The official bilingual statement is included as `conjecture.md`, checked at repository commit `cce2d90cb298de9595798eb674d2c80712100532`.